



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	General Microbiology	PMB101

UNIT – I: History and scope of Microbiology. Recent trends and developments in modern microbiology. Identification, characterization and classification of microorganisms Principles of bacterial taxonomy and classification: - Bergey's manual and its importance, Concepts, nomenclature and taxonomic ranks: general properties of bacterial groups. Major characteristics used in Taxonomy-morphological, physiological and metabolic, ecological, numerical taxonomy, genetic and molecular classification systems; the kingdoms of organisms and phylogenetic trees. Distinguishing characteristics between prokaryotic and eukaryotic cells Structure and function of Cell wall of bacteria, cell membranes, flagella, pili, capsule, gas vesicles, carboxysomes, magnetosomes and phycobiosomes.

UNIT- II: Methods of sterilization: Physical methods – Dry heat, moist heat, radiation methods, filtration methods, chemical methods and their application. Concept of containment facility, sterilization at industrial level. Microbial cultures: Concept of pure culture, Methods of pure culture isolation, Enrichment culturing techniques, single cell isolation, and pure culture development. Microscopic identification characteristics, staining methods – simple staining, differential staining, structural staining and special staining methods.

UNIT -III: Microbiological media-Natural and synthetic; autotrophic, heterotrophic and phototrophic media : basal, defined, complex, enrichment, selective, differential, maintenance and transport media Preservation and Maintenance of Microbial cultures: Repeated sub culturing, preservation at low temperature, sterile soil preservation, mineral oil preservation, deepfreezing and liquid nitrogen preservation, drying, glycerol cultures, freeze drying (lyophilization).

UNIT -IV: Bacterial nutrition and growth kinetics- synchronous, stock, batch and continuous cultures. Growth measurement methods –Metabolic diversity, measurements of NAD, ATP, DNA, and Protein, CO₂ liberated O₂ consumed, extra cellular enzymes. Cultivation of aerobes and anaerobes. Reproduction and spore formation in bacteria. Morphology, Ultra structure and chemical composition of bacteria, actinomycetes, spirochetes, rickettsiae, mycoplasma, Chlamydiae.



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Microbial Physiology and Metabolism	PMB102

UNIT I. Growth and cell division: Measurement of growth, growth physiology, cell division, growth yields, growth kinetics, steady state growth and continuous growth.

UNIT II. Solute Transport: Primary and Secondary transport: Introduction, Kinetics, ABC transporters, Phosphotransferase system, Drug export systems, amino acid transport.

UNIT III. Central Metabolic Pathways and Regulation: Glycolysis, PPP, ED pathway, Citric acid cycle: Branched TCA and Reverse TCA, glyoxylate cycle.

UNIT IV. Protein and Nitrogen metabolism: Metabolism of amino acids: Amino acid biosynthesis and utilisation. Catabolism of amino acid, transamination, decarboxylation and oxidative deamination.

UNIT V. Metabolism of lipids and hydrocarbons: Lipid composition of microorganisms, biosynthesis and degradation of lipids.

UNIT VI. Metabolism of nucleotides: Purine and pyrimidine biosynthesis, regulation of purine and pyrimidine biosynthesis, inhibitors of nucleotide synthesis.

UNIT VII. Physiological Adaptations and Intercellular signaling: Introduction to two component system, regulatory systems during aerobic- anaerobic shifts: Arc, Fnr, Nar, FhlA, regulon, response to phosphate supply: The Pho regulon. Quorum sensing: A and C signaling system, sporulation in *Bacillus subtilis*, control of competence in *Bacillus subtilis*. Heat-Shock responses, pH homeostasis, osmotic homeostasis.



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Virology	PMB103

UNIT I. Animal Viruses: Classification, Morphology and Chemistry of Viruses:

Virus evolution and classification, properties of viruses, virus structure. Techniques for visualisation and enumeration of viral particles, measuring biological activity of viruses, characterization of viral products expressed in infected cells, Diagnostic virology, Physical and chemical manipulation of viruses.

UNIT II. Virus replication Strategies:

Principal events involved in replication: Adsorption, penetration, uncoating nucleic acid and protein synthesis, intracellular trafficking, assembly, maturation and release, viral-host interaction, Host response to viral infection. Replicative strategies employed by animal DNA viruses. Replicative strategies employed by animal RNA viruses. Identification of virus prototypes associated with different virus replication schemes; Details on important viruses namely Herpesvirus, Poliovirus, Influenza virus, VSV, SV40 and Adeno Virus, Poxviruses, Hepatitis Viruses, coronaviruses, Retroviruses. Subviral pathogens: HDV, Prions, Viroids.

UNIT III. Pathogenesis of viral infection and control of viral diseases:

Stages of infection, Patterns of some viral diseases- epidemiology, transmission, infection, symptoms, risk, transformation and oncogenesis, emerging viruses. Host specific and nonspecific defense mechanisms involved in resistance to and recovery from virus infections. Role of interferon in viral infections. Viral Chemotherapy: Nucleoside analogs, reverse transcriptase inhibitors, protease inhibitors, History of vaccines especially smallpox and polio. New methods: subunit vaccines, anti-idiotypic and DNA vaccines.

UNIT IV. Plant and microbial viruses:

General methods of propagation of plant viruses; purification of plant viruses using centrifugation, chromatography and electrophoresis techniques, their assay and comparison of the sensitivity of assay methods; methods employed in identification of plant viruses and structural and functional genomics.

UNIT V. Symptoms of plant virus diseases, transmission of plant viruses, viral and viroid diseases and their control:

General discussion on symptoms caused by viruses and viroids in diseased economically important trees and agricultural crops, and their control including development of virus disease resistant transgenics.



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Industrial Microbiology and Bioprocess Engineering	PMB104

UNIT I.Introduction to industrial microbiology: Sources of industrially important microbes, strain development,types of fermentation and fermenters, process optimization, and recent developments in fermentation technology.

UNIT II.Downstream processing of microbial products: Filtration, centrifugation, cell disruption, liquid-liquid extraction,chromatography, membraneprocesses, drying (lyophilisation and spray drying), and crystallization.

UNIT III.Fermentation economics: Basic objective for successful economically viable fermentation process, cost breakdown for well established fermentation processes, market potential of the products, cost aspects of various stages intheprocesses development including effluenttreatment.

UNIT IV.Production aspects: Microbial strains, substrates, strain improvement, flowdiagrams, product optimization,and applications of industrial alcohol (ethanol and butanol), amino acids (lysine, phenylalanine, tryptophan),antibiotics (cephalosporins, tetracyclines, polyenes), enzymes and immobilized enzymes, SCP, microbial polyesters,biosurfactants, and recombinant products (insulin, somatostatin, thaumatin).

UNIT V.Bioprocessing Technology and Bioengineering: An introduction to fermentation processes- range of fermentation process, microbial biomass. Microbial growth kinetics- batch Culture, continuous culture, industrial application of continuous culture processes, fed- batch culture. The isolation, preservation and improvement of industrially important and usefulmicroorganism.

UNIT VI.Industrialfermentation-typical media, media formulation, water, energy and carbon sources, nitrogen sources, minerals.Vitamin sources, nutrient recycle, buffers, precursors and metabolic regulators, oxygen requirements.Media sterilization, sterilization of fermenter, sterilisationof the feed.Inocula for industrial fermentation- development of inoculafor yeast, bacteria,fungi and actinomycetes, the inoculation of fermenters.Designof fermenter, basic functions, construction, aeration and agitation, oxygen requirements of industrialfermentation.



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Practical – I Lab	PMB155

1. Study of typical compound microscope
2. Micrometry measurement of a fungal spore
3. Determination of cell density by counting chamber
4. Isolation of bacteria from soil, air and water
5. Isolation of fungi from soil
6. Contact slide technique
7. Streak and Pour plate techniques
8. Simple staining of a bacterium
9. Negative staining of a bacterium
10. Gram staining — Positive and negative
11. Separation of DNA by agarose gel electrophoresis -
- 12) Estimation of RNA by Orcinol method.
- 13) Separation of amino acids by paper chromatography.
- 14) Separation of serum proteins by paper electrophoresis.
- 15) Thin layer chromatography of mycotoxins
- 16) SDS-Page of proteins.
- 17) Performance of affinity chromatography.
- 18) Performance of Gel filtration chromatography.
- 19) Demonstration of blotting technique.[any one].
- 20) Ion exchange chromatography



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Cell Biology and Analytical Techniques	PMB201

UNIT I.Organellar Biology: structure, function and biogenesis of chloroplast and mitochondria, mesosomes, lysosomes and cytoskeletal system. Photosynthesis in bacteria and plants, oxygenic and anoxygenic photosynthesis, PSI and PSII, electron transport, CO₂ fixation, purple green and halo bacteria photosynthesis. Physiochemical properties of bacteria- intracellular osmotic pressure, permeability of bacterial cell, nutrient transportsimple diffusion, active, passive and facilitated diffusion.

UNIT II. Signal transduction in eukaryotes: Protein kinases, phosphorylation cascades, Ras pathway, MAP kinase pathway, cyclic nucleotides, G proteins.

UNIT III. Microscopy- Basic principles and application of light, phase contrast microscopy, fluorescent and electron microscope- scanning and transmission. Microtome and sample preparations- fixing of specimens, preparation of block, staining of biological samples. Principles of cytometry and flow cytometry.

UNIT V. Analytical Techniques: Principles of centrifugation,techniques, preparative and analytical methods, density gradient centrifugation. General principle and application of chromatography-paper, column, thin layer, Gas, Ion Exchange, affinity chromatography, HPLC and Gel filtration.Electrophoresis-moving boundary, zone electrophoresis, immunoelectrophoresis, immunoblotting, isoelectric focussing, 2-D electrophoresis.

UNIT VI. Principles, laws of absorption and radiation. Visible, ultraviolet, infrared and mass spectrophotometry. Absorption spectra, flame photometry, NMR,ESR, principles of colorimetry, turbidometry, viscometry. Determination of size, shape and molecular weight of macromolecule- light scattering, diffusion, sedimentation, optical rotatory dispersion and X ray diffraction.

UNIT VII. Radio isotopic tracers- methodology, radiometric analysis, stable and radioactive isotopes, preparation, labelling, detection and measurement of isotopes. RIA, Kinetics of radioactive disintegration, manometric techniques, freeze drying and its application in biologicalsystems



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Bio-Molecules and Enzymes	PMB202

UNIT I. Major Biomolecules: Carbohydrates – Classification, chemistry, properties, and function –. Conjugated polysaccharides– lycoproteins, muriens and lipopolysaccharides.

UNIT II. Lipids – classification, chemistry, properties and function –Conjugated lipids – lipoproteins. Major steroids of biological importance – prostaglandins.

UNIT III. Amino acids and proteins: classification, structure and function. Peptide structure. Ramachandran's plot.. Structural levels of proteins – primary, secondary, tertiary and quaternary, denaturation of proteins. Hydrolysis of proteins, Protein sequencing using various methods.

UNIT IV. Nucleic acids – Structure, function and their properties. Structural polymorphism of DNA, RNA. Structural characteristics of RNA. Sources,.

UNIT V. Enzymology- Introduction, General characteristics of enzymes, Activation energy, Coupled reactions, active site and its importance, Factors influencing catalytic efficiency. Enzyme kinetics, Rapid Equilibrium, Henry-Nucgaekkus-Menten's equations, Steady State approach, significance of Km, Haldane equation, Velocity vital Substrate concentration curves. Methods of plotting enzyme kinetics data-Lineweaver-Burk. Equilibrium dialysis, Effect of pH and temperature on enzyme stability and activity, Arrhenius equation.

UNIT VI. Regulation of enzyme activity: Feedback inhibition, reversible covalent modification, irreversible covalent modification, allosteric concept, Aspartate transcarbamylase, ligand-protein interaction, scatchard plot, Hill plot, cooperativity index, Models for allostery (MWC, KNF), Half site reactivity. Enzyme Inhibition, Models and types of inhibition.

UNIT VII. Applied enzymology: Application of enzymes in analytical labs. (clinical and industrial), enzymes as industrial catalysts, Immobilized enzymes, enzyme electrodes, assay of enzyme activities for diagnostic purposes, abzymes, recent developments



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Environmental Microbiology	PMB203

Unit 1 Brief history and development of environmental microbiology: History and development of microbial ecology highlighting significant contributions of microbiologists and emergence of environmental microbiology, and significant applications of microbes in solving environmental pollution problems.

Unit – 2 Environment and Ecosystems Definitions, biotic and abiotic environment. Environmental segments. Composition and structure of environment. Concept of biosphere, communities and ecosystems. Ecosystem characteristics, structure and function. Food chains, food webs and trophic structures. Ecological pyramids.

Unit – 3 Eutrophication Water pollution and its control, Eutrophication, causes of eutrophication, effects of eutrophication on the quality of water environment, factors influencing eutrophication. Qualitative characteristics and properties of eutrophic lakes. Algae in eutrophication, algal blooms, their effects and toxicity, coloured waters, red tides, and cultural eutrophication. Physico-chemical and biological measures to control eutrophication

Unit –4 Effluent treatment techniques Microbiology of wastewater and solid waste treatment: - Waste-types-solid and liquid waste Characterization, physical, chemical, biological, aerobic, anaerobic, primary, secondary and Tertiary treatments. Anaerobic processes: Anaerobic digestion, anaerobic filters, and up flow anaerobic sludge. Treatment schemes for effluents of dairy, distillery, tannery, sugar and antibiotic industries. Bioconversion of Solid Waste and utilization as fertilizer. Bioaccumulation of heavy metal ions from industrial effluents.

Unit 5 Biodegradation Factors affecting biodegradation, effects of pesticides, biodegradation of pesticides, mechanism of biodegradation, microorganisms involved, biodegradation of other toxic chemicals. Bioplastics.

Unit – 6 Bioremediation of Xenobiotics Microbiology of degradation of xenobiotics in the environment, ecological considerations, decay behaviour, bioaccumulation and biomagnification, oil pollution, surfactants and pesticides. Genetically Modified Organisms released and its environmental impact assessment and ethical issues. Bioremediation of Petroleum hydrocarbons.

Unit – 7 Global environmental problems Ozone depletion, UV-B, global warming and its impact, ozone layer-formation and depletion, greenhouse effect and acid rain, their impact and biotechnological approaches for management.



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Bio Informatics	PMB204

UNIT I. Introduction to computers and bioinformatics - Types of operating systems, concepts of networking and remoteloin, basic fundamentals of working with unix.

UNIT II. Biological databases- Overview, modes of database search , mode of data storage (Flat file format, db-tables), flatfileformats of GenBank, EMBL, DDBJ, PDB.

UNIT III. Sequence alignment —Concept of local and global sequence alignment, Pairwise sequence alignment, scoring analignment, substitution matrices, multiple sequence alignment. **UNIT IV. Phylogenetic analysis-** Basic concepts of phylogenetic analysis, rooted/uprooted trees, approaches for phylogenetic tree construction (UPGMA, Neighbour joining, Maximum parsimony, Maximum likelihood).

UNIT VI. Generation and analysis of high throughput sequence data - Assembly pipeline for clustering of HTGS data, format of “.ace” file, quality assessment of genomic assemblies, International norms for sequence data quality, Clustering of EST sequences, concept of Unigene. Annotation procedures for high through-put sequence data- Identification of various genomic elements (protein coding genes, repeat elements, strategies for annotation of whole genome, functional annotation of EST clusters, gene ontology (GO) consortium.

UNIT VII. Structure predictions for nucleic acids and proteins - Approaches for the prediction of RNA secondary and tertiary predictions, energy minimization and base covariance models, Basic approaches for protein structure predictions, comparative modeling, fold recognition/“threading” and ab-initio prediction.



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M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Practical – II Lab	PMB255

1. Estimation of proteins by Biuret method and FolinCiocalteau method.
2. Production of citric acid by A.niger. Recovery & Fermentation.
3. Estimation of Ethanol by dichromate method.
4. Production of Ethanol by fermentation and recovery.
5. Preparation of Wine from grapes by fermentation.
6. Production of glutamic acid by fermentation.
7. Estimation of bacteria, actinomyces and fungi in soil by dilution — Plating method.
8. 8 Isolation and enumeration of major groups of microorganisms from rhizosphere and nonrhizosphere.
9. Study of root nodules and isolation of Rhizobium from legume root nodules.
10. Isolation of Azospirillum / Azotobacter.
11. Staining and observation of vesicular-arbuscularmycorrhizal (V AM) fungi.
12. 12.Isolation of microorganisms of air by Petri plate exposure method.
13. Computer operations-getting acquainted with different parts of Computers. [DOS] and basics of operating a computer.
14. Creating files, folders and directories.
15. Applications of computers in biology using MS-Office.
A] MS-Word B] Excel C] Power Point
16. Creating an e-mail account, sending and receiving mails.
An introduction to INTERNET, search engines, websites, browsing and Downloading



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Genetic Engineering	PMB301

UNIT I. The nature of Genetic material: The structure of DNA and RNA; Melting of DNA, Superhelicity, Organization of Microbial Genomes, Organization of Eukaryotic Genomes, Chromatin arrangement, nucleosome formation.

UNIT II. DNA replication: Arrangement of replicons in a genome, Various modes of replication, specific features of replication in Prokaryotes and Eukaryotes, action of topoisomerases, Telomere maintenance and Chromatin Assembly, Single stranded DNA replication. DNA repair and recombination, DNA Mismatch Repair, Double Strand Break Repair, Recombination as a molecular biology tool.

UNIT II. Transcription: Transcription machinery of prokaryotes, eukaryotes, various forms of RNA polymerase promoters, enhancers, silencers, activators, effect of chromatin structure, regulation of transcription.

UNIT III. Post-transcriptional processes: RNA processing, splicing, capping and polyadenylation, rRNA and Trna processing, RNA Editing; RNAi and miRNAs, Antisense RNA, Post-transcriptional gene regulation.

UNIT IV. Translation: The genetic code and protein structure, Mechanisms of translation in prokaryotes, Mechanisms of translation in eukaryotes, *in vi*/rotational systems, polycistronic/ monocistronic synthesis, Regulation of translation, RNA instability, inhibitors of translation, stringent response in bacteria. Post-translational processes: Protein modification, folding, chaperones, transportation; The Signal Hypothesis, protein degradation.

UNIT V. Molecular basis of cell physiology: Signals and cascades in organism development Molecular mechanisms of Oncogenesis and cancer, genetic disorders, aging, mitochondrial inheritance. Implications of genome organization, Genes and behavior, Genome analysis, DNA typing, Genomics and beyond.

UNIT VI. Polymerase Chain Reaction: Concept of PCR and various thermophilic enzymes used in PCR. Gradient PCR versus Touchdown PCR. Designing primers. Long PCR, Inverse PCR, RT-PCR, 5' and 3' RACE, qPCR, Real Time PCR using SYBR Green, Scorpion primers and TaqMan probes, MOPAC, Multiplex PCR, Differential Display PCR, RAPD fingerprinting of micro-organisms.

UNIT VII. . Genome sequencing: DNA sequencing by Sanger's method — traditional and cycle sequencing. Physical mapping by restriction fragment fingerprinting of BAC clones. Whole genome shotgun sequencing. Clone-by-clone shotgun sequencing of genome — preparation of BAC/YAC library, map construction, random shotgun phase, finishing phase and sequence authentication. Genome annotation at the nucleotide level, protein level and process level. Comparative genome sequencing of micro-organisms to identify and categorize SNPs. Array CGH.



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Medical Microbiology	PMB302

UNIT I.General topics on medical microbiology: History and development, Koch's postulates, classification of medically important bacteria. Infection: source, modes of transmission, portal of entry into susceptible host and prevention. Bacterial pathogenicity, identification of bacteria: staining methods, culture method, biochemical tests and other recent methods. Sterilization and disinfection. Normal microbial flora, antimicrobial agents, drug resistance and drug sensitivity test.

UNIT II.Systematic Microbiology: Diseases caused by Gram positive cocci-sore throat, pneumonia etc., Diseases caused by Gram negative cocci- meningitis, gonorrhoea etc. Diseases caused by Gram positive bacilli- Tuberculosis, Diphtheria, Tetanus, Gas gangrene etc, Diseases caused by Gram negative bacilli of Enterobacteriaceae- Enteric fever, Bacillary dysentery, UTI
Diseases caused by Gram negative bacilli- Cholera, plague, Whooping cough, Wound infection, Septicaemia. Sexually transmitted diseases. Disease caused by mycoplasma, Chlamydia, Rickettsia.

UNIT III.Overview of medical Mycology: Important fungal diseases- Superficial, Subcutaneous, Systemic and opportunistic Mycosis. Overview of Medical Parasitology, Important Protozoan Diseases-Ascaris, Ankylostomiasis, Filariasis, Taeniasis, Echinococcosis etc. Overview of Medical Virology, Important Viral Diseases- Herpesvirus, Poliovirus, Rabies Virus, Arboviruses, Hepatitis, HIV etc. Opportunistic Microbial Infection, Water, Milk and Food borne Diseases, Microbial Vaccine.

UNIT IV.Quality Control: Introduction, Total Quality Management, Framework, Laboratory Processes, Assurance and Assessment, Quality control Planning and Improvement.

UNIT V.Haematology: Basic Haematological Disorders- Classification of Anemia, Iron Deficiency anemia, Megaloblastic Anemia, Haemolytic Anemia, Basic Haematological Techniques- Collection of Blood Specimens, Haemolysis of Blood, Separation of Serum and Plasma, Maintenance and Transport of Specimen, Coagulation and Bleeding Disorders in brief).



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Agricultural Microbiology	PMB303

UNIT I. Soil Microorganism in agro ecosystem: Types of microbial communities: soil microbial diversity: significance and conservation; effect of agricultural practices on soil organism..

UNIT II. Biological Nitrogen fixation: The range of nitrogen fixing organisms; mechanism of nitrogen fixation (biochemistry of nitrogenase); genetics of nitrogen fixation. Rhizobium Legume Association; Symbioses, N₂ fixation by non leguminous plants.

UNIT III. Chemical transformation of microbes: Organic matter decomposition, nutrient mineralization and immobilization; transformation of carbon and carbon compounds.

UNIT IV. Biodegradation of herbicides and pesticides

UNIT V. Biofertilizers: Mass cultivation of microbial inoculants; green manuring; algalization; Azolla.

UNIT VI. Microbial products and plant health: Plant growth promoting rhizobacteria (PGPR); significance of mycorrhizae, Microbial herbicides, biological control.



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Immunology	PMB304

UNIT I.ImmuneSystem:Threefundamental concepts : Specificity, discrimination of self from non-self and memory.Lymphocytes, theirsubpopulation, properties and functions, lymphocyte trafficking.

UNIT II.Antigens and Immunoglobulins

Concept of haptens, determinants, conditions of antigenicity, antigens and immunogenicity, superantigen.Immunoglobulins: Structure and properties of immunoglobulin classes.Theories of antibody formation, hybridomatechnology for monoclonal antibodies and designer monoclonal antibodies.Multiplemyelomas and structural basis of antibody diversity.Freund's adjuvants anditssignificance.

UNIT III.Genetic organization: Organization of the genes for B and T cell receptors. Genetic organization of MHC-I andMHC-II complex (both HLA and H-2).Peptide loading and expression of MHC-I and MHC-II molecules.

UNIT IV.Immune response and signaling: Humoral and cell- mediated immune response; Innate immune response andpattern recognition; Recent advances in innate immune response especially NK-DC interactions; Major cytokinesand their role in immune mechanisms: TNF, IFN, IL-1, IL-2, IL-4, IL-6, IL-10, IL-12, IL-17, TGF β ; Cell signalling through MAP kinases and NF- κ B.

UNIT V.Immunity and Immunoassays: Defenseagainst bacteria, viruses, fungi and parasites. Immunodiagnostics and immunotherapy invirology — Serological methods for detection and quantitation of viruses including Hepatitis,Influenza, HIV and others.Immuno- assays: ELISA, ELISA-PCR, RIA, Western Blotting, Immunofluorescensandtheir application. Tolerance and autoimmunity:Centraland peripheral tolerance, and their mechanism; Mechanisms ofautoimmunity; Autoimmune components of diabetes mellitus (DM), multiple sclerosis (MS), experimentalautoimmune encephalitis (EAE); Infections leading to autoimmune diseases.

UNIT VI.Immunological disordersand hypersensitivity: Deficiencies / defects of T cells, B cells, complement andphagocyticcells;immunodeficiency with special reference to AIDS, Comparative study of Type I-V hypersensitivities with examples.Mechanism and molecular events in mast cell degranulation by IgE,pharmacological mediators of type-I reactions.

UNIT VII.Transplantation and tumor immunology: Alloreactive response; types of grafts, Graft rejection and GVHD; HLA-matching; mechanismand prevention of graft rejection.Transgenic animals for xenotransplantation; Tumor antigens, immune response to tumors and immunotherapy oftumors.



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M.Sc. (Microbiology)

UNIT VIII.Vaccine:Vaccines — Introduction and History, Effectiveness and Adverse effect of Vaccines,Types of Vaccines, Production of Vaccines, Delivery system of Vaccines, Hepatitis B, AIDS, and DNA vaccines, DIVA Vaccines,Recent Advances in Vaccines,Vaccines for Cancer.



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Practical —III Lab	PMB355

i. Plasmid Curing by AcridineOrange

2 Restriction Digestion of Z DNA using three Restriction Endonucleaseenzymes:

a) EcoR V b) Hind III c) BamH I

3. Replica platingtechniques

4. Agarose gel electrophoresis and restriction mapping ofDNA

5. Demonstration of techniques ofPCR

6. Isolation of Genomic DNA from bacterial cell / plantcell

7. Isolation of RNA from Yeastcells

8. Determination of T_m values ofDNA

9. Isolation of Temperature sensitive conditionalMutant

10. Isolation of auxotrophic mutant by 5BrUmutagenesis

11.BacterialConjunction

12.To perform test for antibiotics sensitivity by disc method. To determine the minimum inhibitory concentration of given antibiotics.

1. Preparation of blood smear.

2. To isolate serum from blood plasma.

3. To perform agglutination reaction to identification of blood group.



RKDF UNIVERSITY, RANCHI

M.Sc. (Microbiology)

Branch	Subject title	Subject Code
M.Sc.(Microbiology)	Dissertation/ Project Work	PMB401

Project model (if any) and the Project record notebook = 500
dissertation work may be as per the following guidelines:

Project model (if any) and the Project record notebook = 500

Project presentation and viva-voce = 100

Overall project dissertation may be evaluated under the following heads:

- ☐ Motivation for the choice of topic
- ☐ Project dissertation design
- ☐ Methodology and Content depth
- ☐ Results and Discussion
- ☐ Future Scope & References
- ☐ Presentation style
- ☐ Viva-voce

PROJECT WORK/ DISSERTATION/ PAPER PRESENTATION

Each student has to submit two copies of the dissertation work duly forwarded by the HOD of Department concerned. The forwarded copies will be submitted in the Department of Microbiology, RKDF University, for evaluation (Seven days before the seminar).

The paper will consist of

- (a) Field work/Lab work related to the project.
 - (b) Preparation of dissertation based on the work undertaken.
 - (c) Presentation of project work in the seminar on the assigned topic in the P.G. Department of Microbiology, RKDF University, Ranchi & open viva thereon.
- ☐ Student alone or in a group of not more than five, shall undertake one Project approved by the Subject Teacher/H.O.D. of the Department/College concerned. The progress of the Project shall be monitored by the faculty members at regular intervals